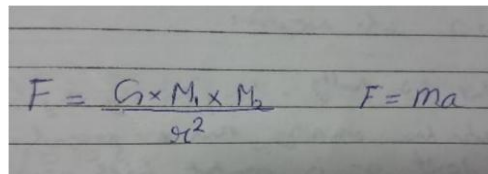


Orbital Simulation using Newtonian Mechanics

1. Introduction

For a long time, there was a misconception about the planetary motion of our solar system that Earth is the center of the solar system and all other planets revolve around it. Then scientists like Kepler and Newton came and explained the truth with the help of mathematics, providing laws of physics that helped people understand how our solar system actually works.

However, for a long time, people did not truly believe these ideas, as they could not directly observe such motion with their own eyes. This was something even I found hard to believe when I first learned about Newton's formulas:



A photograph of a piece of lined paper with two physics formulas written in blue ink. The first formula on the left is $F = \frac{G \times M_1 \times M_2}{r^2}$, representing the law of universal gravitation. The second formula on the right is $F = ma$, representing Newton's second law of motion.

I couldn't relate how this formula causes the Earth to revolve around the Sun, or how different planets revolve at different distances and speeds. Why does Earth take 365 days to complete one revolution, while other planets take more or less time?

These were some of the questions and doubts that motivated me to make this project.

As we know, studying celestial bodies is both fascinating and important to understand how our planet works, how day and night occur, how seasons change, how tides form in the ocean, and more. It's also essential to understand other planets and to predict their positions for purposes like satellite launching.

So, in this project, we aim to simulate the motion of Earth and Jupiter around the Sun in space using Newton's laws. By doing this, we can understand how planetary motion is affected by gravitational forces and how trajectories change over time.

We aim to:

- Visualize planetary orbits using Python and Matplotlib animation
- Visualize planetary motion based on gravity and initial conditions
- Explore orbital periods, distances, and the effects of velocity
- Understand numerical simulation through Euler integration
- Understand gravitational force-based motion using fundamental physics laws

This helps build both a physical understanding of space systems and a computational approach to simulating real-world mechanics.

2. Model Formulation (Equations / Rules)

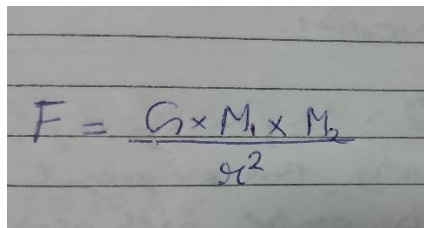
In our model, we used the basic laws of Newton and very simple mathematics to keep things straightforward and easy to understand. We avoided using complex calculus, but we did apply **Euler integration** to calculate the position and velocity of the planets — which is a simple and intuitive numerical method.

We have based our simulation on:

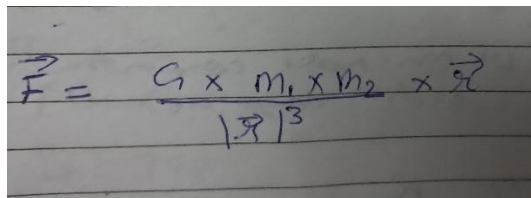
- **Newton's Universal Law of Gravitation**
- **Newton's Second Law of Motion**
- **Euler Integration** for motion updates

Also, the values we used — like the mass of Earth, Jupiter, and the Sun, as well as their velocities — are not exact, but they are very close to the real values of these celestial bodies.

Gravitational Force Equation


$$F = \frac{G \times M_1 \times M_2}{r^2}$$

Since we are using vectors (which is important in planetary motion), the formula becomes:

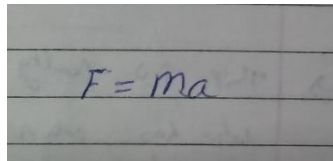

$$\vec{F} = \frac{G \times m_1 \times m_2}{|r|^3} \times \vec{r}$$

Where:

- G is the gravitational constant: 6.67430×10^{-11}
- m_1, m_2 are the masses of the two bodies between which the force is calculated
- r is the **distance** between the two bodies
- $\vec{r}$ is the **position vector** from one body to the other, showing the **direction** of force

This gravitational force acts as a vector in the direction of the other body.

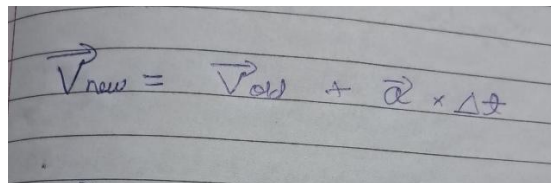
Acceleration Calculation


$$F = ma$$

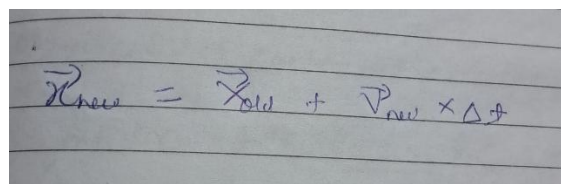
Motion Equations: Euler Integration

We used the **Euler method**, which is not very accurate compared to higher-order methods, but it is simple and fast. Normally, we would need calculus to compute exact motion, but here we use this numerical approach to update values step-by-step.

1. Velocity Update


$$\vec{V}_{new} = \vec{V}_{old} + \vec{a} \times \Delta t$$

2. Position Update


$$\vec{r}_{new} = \vec{r}_{old} + \vec{V}_{new} \times \Delta t$$

Using these equations, we update the position of each planet at every time step.

Assumptions in the Model

1. The Sun is fixed at the origin and exerts gravitational force on both Earth and Jupiter.
 2. Earth and Jupiter do **not** affect each other's motion (simplified two-body interaction).
 3. Time is measured in seconds, and the simulation is in **2D space**.
 4. The Sun and the entire solar system are assumed to be stationary.
 5. The gravitational force acts **along the direction** of the distance vector from the Sun to the planet.
 6. We have given each planet some **initial velocity** so that it begins to revolve around the Sun in an orbit.
-

3. Implementation of the Model in Python

We used **NumPy** and **Matplotlib's animation library** to simulate and visualize our planetary model.

- **NumPy** was used for efficient vector and numerical operations.
- **Matplotlib's FuncAnimation** was used for real-time visualization of the motion of the planets.

1. Initialization

We begin by defining the **masses**, **initial positions**, and **initial velocities** of the celestial bodies:

Masses (in kg):

- **Sun:** 1.989×10^{30}
- **Earth:** 5.972×10^{24}
- **Jupiter:** 1.898×10^{27}

Initial Positions (from Sun):

- **Earth:** ~147 million km (1.47×10^8 km)
- **Jupiter:** ~778 million km (7.78×10^8 km)

Initial Orbital Velocities:

- **Earth:** ~29.78 km/s
- **Jupiter:** ~13.07 km/s

We approximate these values for simplicity, ensuring they are close enough to real astronomical values to reflect realistic orbital motion.

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

G = 6.67430e-11
mass_sun = 1.989e30
mass_earth = 5.972e24
mass_jupiter = 1.898e27

time_step = 60 * 60 * 24
total_steps = 10000

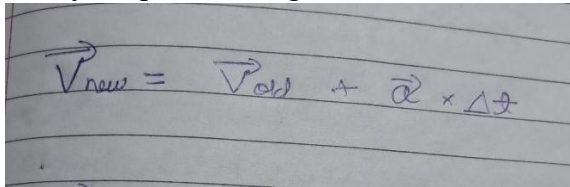
earth_pos = np.array([1.8e11, 0.0])
earth_vel = np.array([0.0, 24000.0])
jupiter_pos = np.array([7.785e11, 0.0])
jupiter_vel = np.array([0.0, 12500.0])

earth_path = []
jupiter_path = []
```

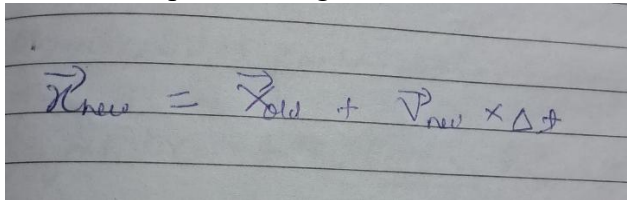
2. Simulation Loop

We then simulate planetary motion using a time-stepping loop. At every step:

- **Gravitational force** from the Sun is calculated for both Earth and Jupiter using Newton's law.
- **Acceleration** is computed using $\vec{a} = \vec{F}/m$
- **Velocity** is updated using Euler's method:


$$\vec{V}_{new} = \vec{V}_{old} + \vec{a} \times \Delta t$$

- **Position** is updated using:


$$\vec{r}_{new} = \vec{r}_{old} + \vec{V}_{new} \times \Delta t$$

- **Positions are stored** in a list to be used later for visualization.

We run this loop for **10000 days**, updating and storing the position of each planet at every step.

```
for step in range(total_steps):

    r_earth = -earth_pos
    distance_earth = np.linalg.norm(r_earth)
    force_earth = G * mass_sun * mass_earth / (distance_earth ** 3) * r_earth
    acceleration_earth = force_earth / mass_earth
    earth_vel = earth_vel + acceleration_earth * time_step
    earth_pos = earth_pos + earth_vel * time_step
    earth_path.append(earth_pos.copy())

    r_jupiter = -jupiter_pos
    distance_jupiter = np.linalg.norm(r_jupiter)
    f_jupiter = G * mass_sun * mass_jupiter / (distance_jupiter ** 3) * r_jupiter
    acc_jupiter = f_jupiter / mass_jupiter
    jupiter_vel = jupiter_vel + acc_jupiter * time_step
    jupiter_pos = jupiter_pos + jupiter_vel * time_step
    jupiter_path.append(jupiter_pos.copy())

earth_x, earth_y = zip(*earth_path)
jupiter_x, jupiter_y = zip(*jupiter_path)

fig, ax = plt.subplots(figsize=(8, 8))
ax.set_facecolor("black")
ax.set_xlim(-9e11, 9e11)
ax.set_ylim(-9e11, 9e11)
ax.set_aspect('equal')
```

3. Visualization

For visualization, we used matplotlib FuncAnimation to create a smooth, real-time animation of the planetary orbits.

- The **Sun** is displayed as a **yellow dot** at the center.
- **Earth** and **Jupiter** move along their orbital paths, leaving **trails** behind to indicate their movement over time.
- The **current simulation time (in days)** is displayed on the screen during the animation.

This approach provided a **clear visual representation** of the differing orbits, velocities, and distances of Earth and Jupiter relative to the Sun.

```
sun = ax.plot(0, 0, 'yo', markersize=12, label='Sun')[0]
earth_dot, = ax.plot([], [], 'bo', label='Earth')
jupiter_dot, = ax.plot([], [], 'orange', marker='o', label='Jupiter')
earth_line, = ax.plot([], [], 'b-', alpha=0.5)
jupiter_line, = ax.plot([], [], 'orange', alpha=0.5)

time_text = ax.text(0.05, 0.95, '', transform=ax.transAxes, color='white', fontsize=12)

ax.legend()

def update(frame):

    earth_dot.set_data([earth_x[frame]], [earth_y[frame]])
    jupiter_dot.set_data([jupiter_x[frame]], [jupiter_y[frame]])

    earth_line.set_data(earth_x[:frame+1], earth_y[:frame+1])
    jupiter_line.set_data(jupiter_x[:frame+1], jupiter_y[:frame+1])

    time_in_days = frame * (time_step / (60 * 60 * 24))
    time_text.set_text(f'Time: {int(time_in_days)} days')

    return earth_dot, jupiter_dot, earth_line, jupiter_line, time_text

ani = FuncAnimation(fig, update, frames=total_steps, interval=20, blit=True)
plt.show()
```

4. Exploration And Observation of the model

In this section, we experimented with different initial parameters to observe how planetary motion is affected.

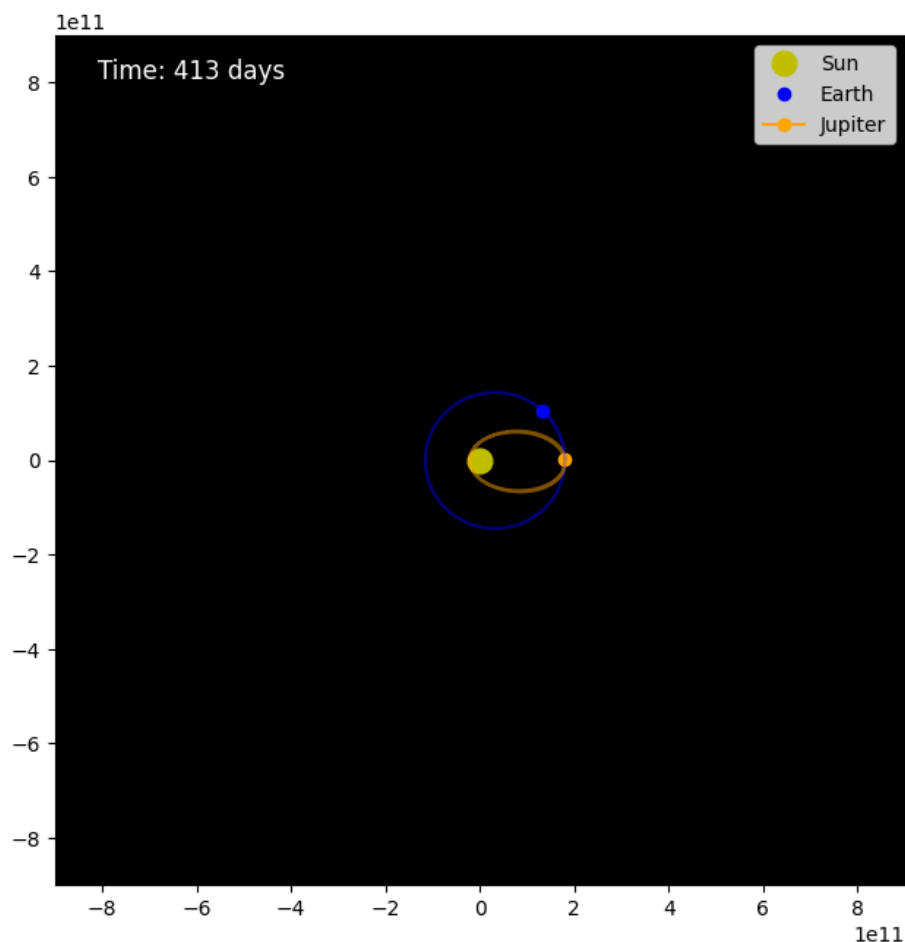
4.1 Same Mass and Position, Different Velocities

- **Initial Mass:** Both Earth and Jupiter set to 5.972×10^{24}
- **Initial Position:** 1.8×10^{11} km from the Sun (same for both planets)
- **Different Velocities:** Earth and Jupiter were given different initial velocities.

Observation:

Despite having the same mass and distance from the Sun, their orbital paths varied significantly due to the difference in velocity. The planet with a higher velocity completed orbits more quickly, while the slower one moved on a larger, slower trajectory.

<https://drive.google.com/file/d/1UKUeQKP-4zs-Wd7vyYae2EzLD2CbeQGf/view?usp=sharing>



4.2 Changing Initial Velocities

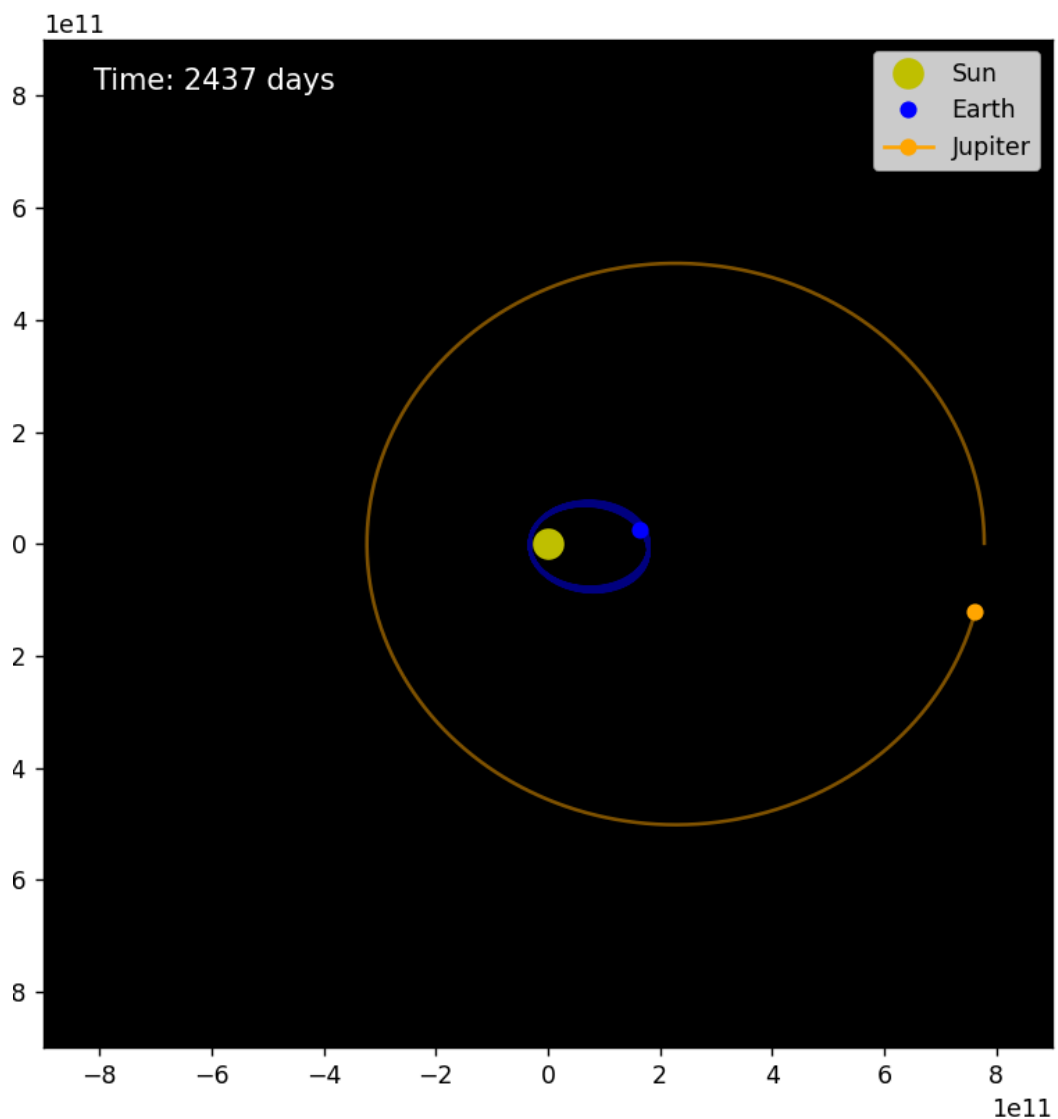
We tested different sets of velocities and recorded their effects on orbital motion:

Moderate Reductions:

- Earth: from 24,000 m/s to 15,000 m/s
- Jupiter: from 12,500 m/s to 10,000 m/s

Observation:

As velocities were lowered, orbits became more elliptical. Earth passed closer to the Sun during its orbit.

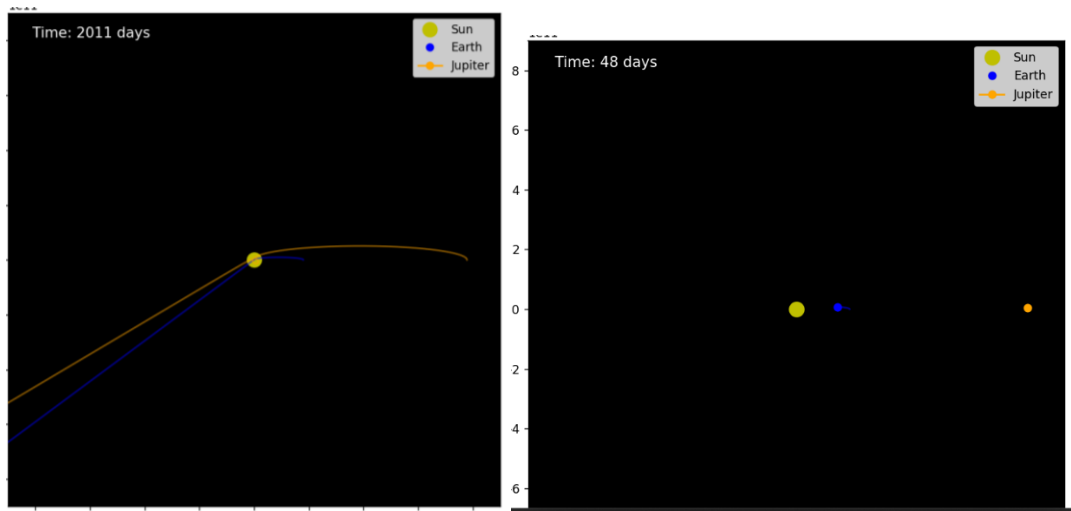


Very Low Velocities:

- Earth: 2,000 m/s
- Jupiter: 1,200 m/s

Observation:

With insufficient velocity, both planets spiraled inward and collapsed into the Sun, unable to maintain stable orbits.

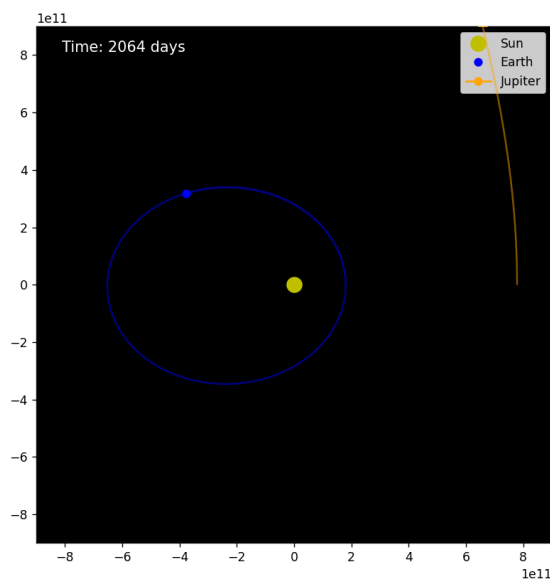


High Velocities:

- Earth: from 24,000 m/s to 34,000 m/s
- Jupiter: from 12,500 m/s to 23,500 m/s

Observation:

Earth formed a highly elliptical orbit, traveling very far from the Sun. Jupiter, with extremely high speed, **escaped the gravitational pull of the Sun**, indicating that its velocity exceeded the **escape velocity** at that distance.

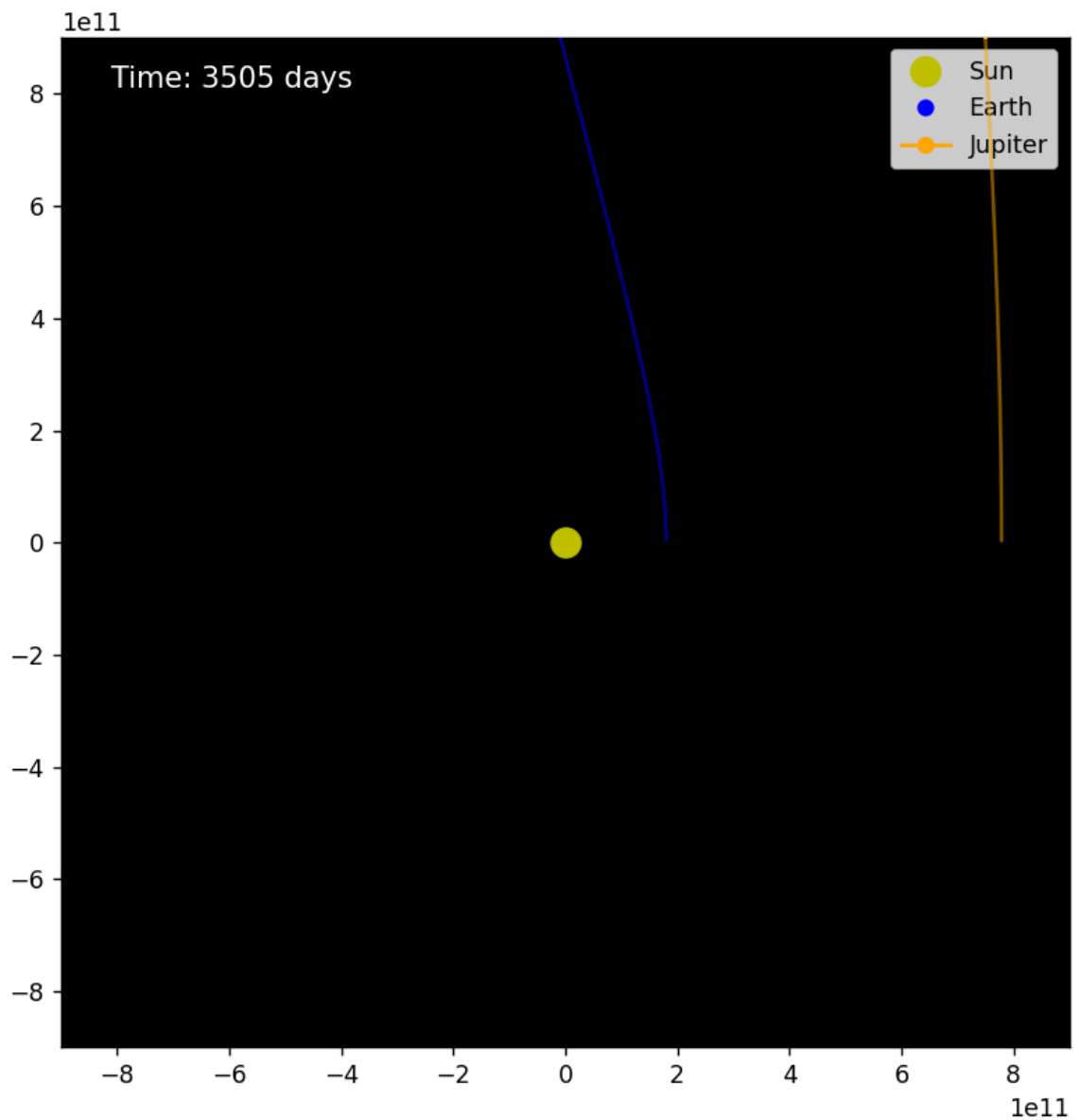


Excessive Velocities:

Both planets were given even higher speeds to **simulate escape orbit**.

Observation:

Planets no longer orbited the Sun but instead followed open, hyperbolic paths — escaping the system entirely.

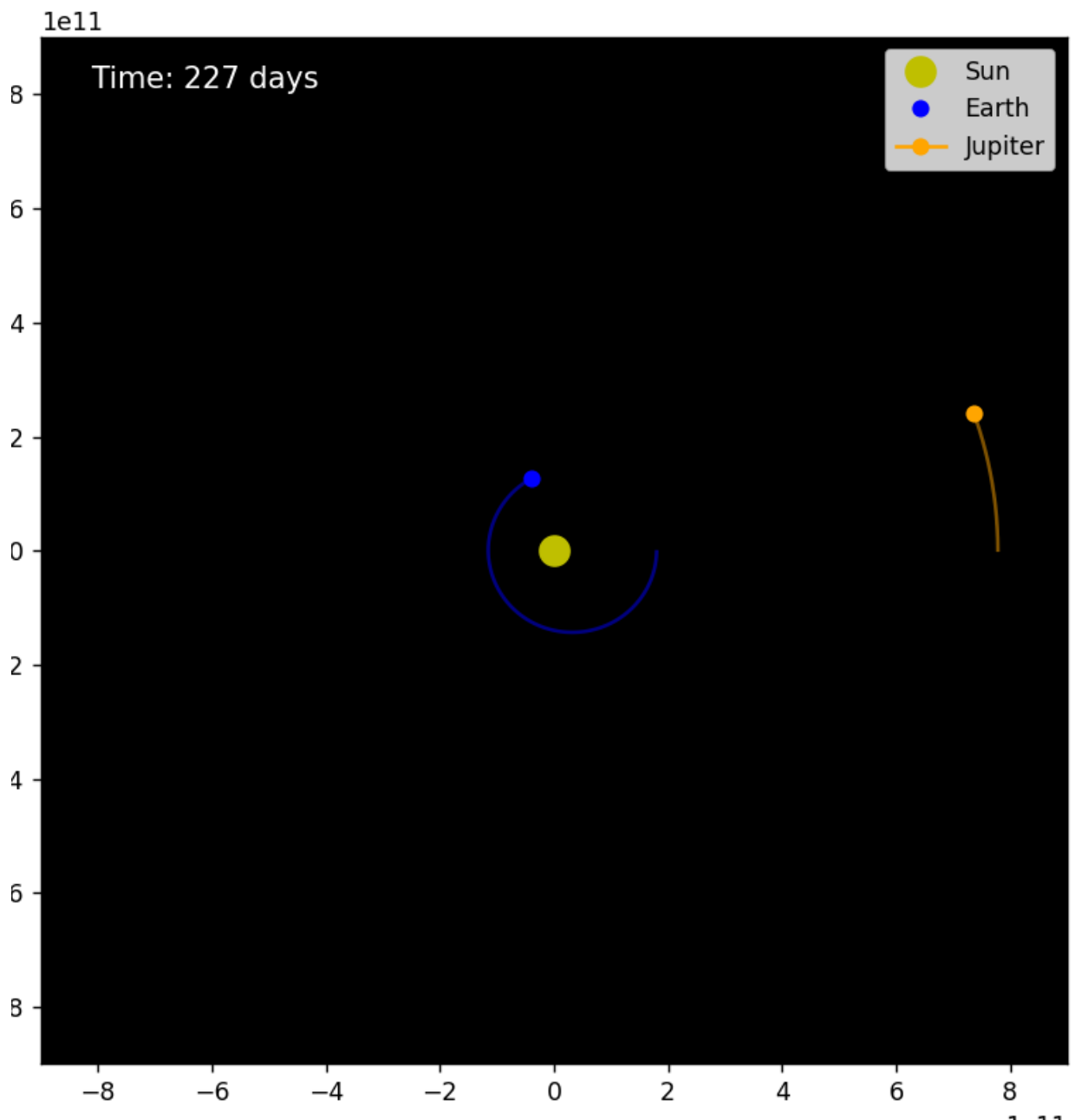


4.3 Negative Initial Velocity

We tested the effect of assigning **negative velocity** (inverting direction) to Earth.

Observation:

Earth's motion reversed direction, orbiting clockwise instead of counterclockwise. This shows that the direction of velocity directly controls the direction of revolution.



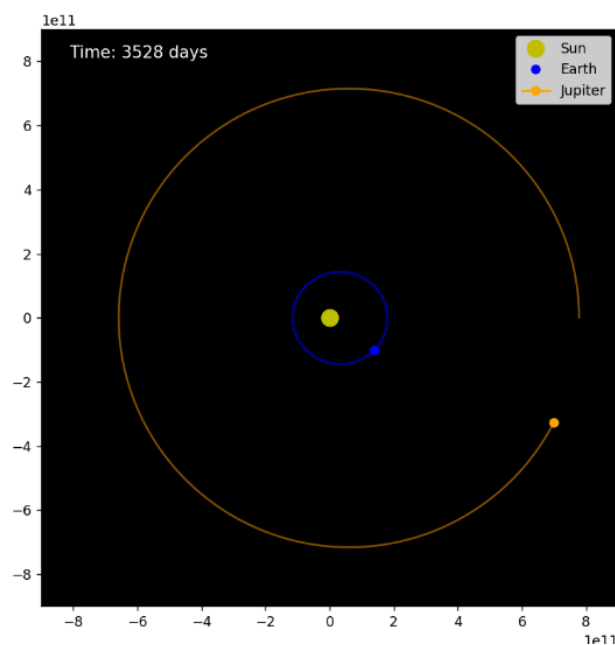
4.4 Changing Planet Mass

- Earth and Jupiter were each given different masses:
 - Earth: from 5.972×10^{24} kg to 1.0×10^{25} kg
 - Jupiter: from 1.898×10^{27} kg to 1.0×10^{24} kg
- Initial position and velocity were kept the same as in earlier simulations.

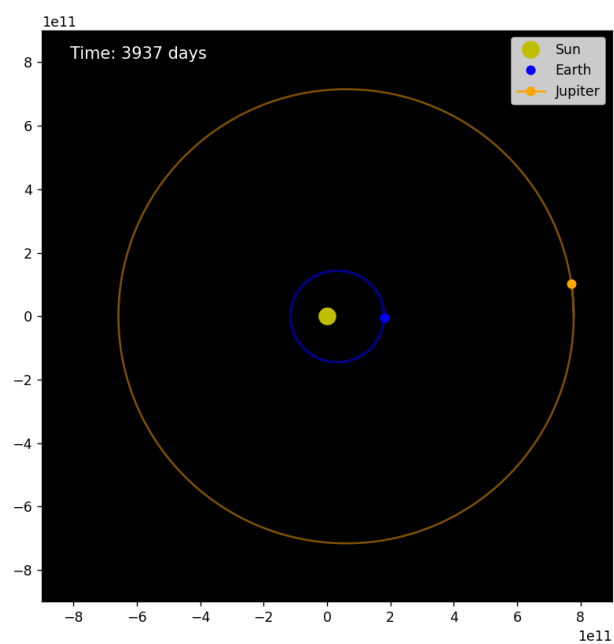
Observation:

No changes can be observed in the moment of the planet and their speed .

a. With original mass of planet



b. With change mass of planet



4.5 Changing Sun's Mass

Sun's mass is changed

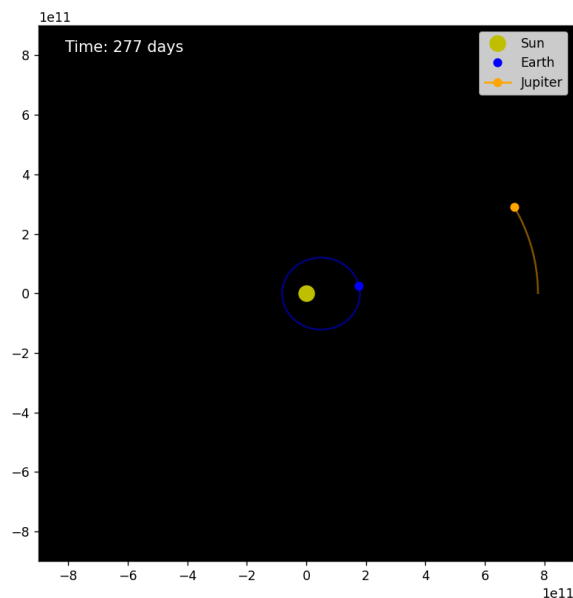
- From 1.989×10^{30} kg (real value) to 2.5×10^{30} kg and then to 1.0×10^{30} kg
- Planet masses, positions, and velocities remained constant.

Observation:

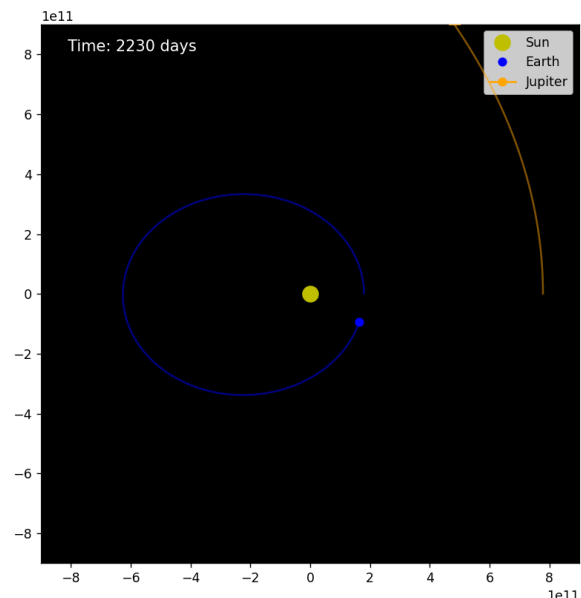
When the mass of the Sun increases, the gravitational pull becomes stronger, causing the planet's speed to increase and its revolution time to decrease — in this case, Earth completes one orbit in nearly 270 days instead of the usual 365.

When the Sun's mass is reduced, Earth's orbital path becomes wider, indicating weaker gravitational attraction. As a result, Jupiter even reaches escape velocity due to the reduced gravitational force.

a. When mass of sun is 2.5×10^{30}



b. When mass of sun is 1.0×10^{30}



Observation : -

- Earth completes nearly 3 orbits in 1000 days. Jupiter barely completes 1/10th, showing a long orbital period.
- Trails represent stable orbits, confirming that gravitational force and velocity balance each other with respect to other variables.
- Changing mass or velocity directly alters orbit shape and speed.
- Euler integration, while useful, shows small numerical drifts — orbits may slowly expand or contract over long periods.
- Different initial velocities of planets help change the speed of the planet when it is near the Sun. For example, at velocity 24000 the orbit looks circular or slightly elliptical, but at 22000 or 20000 it becomes very elliptical, and the planet comes closer to collapsing into the Sun. At 30000 and above, the orbit becomes extremely stretched.
- Earth, being closer to the Sun, completes more revolutions compared to Jupiter in the same simulation time.
- Jupiter's path is wider and slower, resembling its real-life long orbital period.
- The simulation correctly shows curved orbital paths due to gravity, though not perfect ellipses — because of the simplicity of the Euler integration method. This also connects to how seasons are formed.
- The scale difference between orbits is visible, showing the vastness of distances in the outer solar system.
- If we keep the initial velocity low, the planet collapses into the Sun. If the initial velocity is stable, the orbit becomes circular or elliptical. If the velocity is very high, the planet may escape the Sun's orbit or form a very large and highly elliptical orbit.
- Because of the different velocities, one planet may take more or less time than another to complete one revolution.
- Due to the elliptical shape of orbits, planets have seasons — as the intensity of sunlight varies across their path.
- Also, the velocity of a planet changes based on its distance from the Sun: when it is far, the velocity is slower; when it comes near, the velocity increases.
- In some long simulations, the orbit slowly spirals inward or outward — showing the numerical limits of Euler's method and the importance of precise integration over time.
- Orbits near stable velocity values (like 24000) showed more regular patterns, while too high or low values (like 20000 or 30000) led to collapse or escape — highlighting how finely tuned orbital motion needs to be.
- When we change the mass of the planet it does not significantly affect the shape or period of its orbit. This is because, in Newton's law of gravitation and second law of motion, the mass of the orbiting planet cancels out when calculating acceleration. As a result, regardless of whether Earth's or Jupiter's mass is increased or decreased, the orbital path remains largely unchanged. This reinforces the understanding that gravitational acceleration is independent of the planet's mass.

- When we change the mass of the Sun, it has a direct effect on the orbital motion of planets. When the Sun's mass is increased, its gravitational pull strengthens, causing planets to speed up and complete their orbits more quickly — for example, Earth completes a revolution in around 270 days instead of 365. Conversely, when the Sun's mass is reduced, the gravitational attraction weakens. Planets move slower, their orbits widen, and in some cases like Jupiter, they may reach escape velocity and leave the system altogether.

$M_1 \rightarrow \text{Mass of sun}$
 $M_2 \rightarrow \text{Mass of Earth}$

$$F = \frac{G \times M_1 \times M_2}{r^2} \quad \text{--- (1)}$$

$$F = m_2 \times a \quad \text{--- (2)}$$

Equating (1) & (2)

$$m_2 \times a = \frac{G \times M_1 \times M_2}{r^2}$$

$$\boxed{a = \frac{G \times M_1}{r^2}}$$

$\Rightarrow$ So changing the mass of planets not affect the motion or orbit.

But when we change the mass of sun it affect the motion of planet.

5. Conclusion

From this very simple model, by simulating planetary motion with multiple values, we gain an understanding of how gravitational force governs the movement of planets in our solar system and how the Earth revolves around the Sun. We observe why Earth takes approximately 365 days to complete one revolution — due to its mass, the mass of the Sun, and its initial velocity. Since Earth's orbit is elliptical rather than a perfect circle, we also understand how this leads to the changing of seasons.

By observing both Earth and Jupiter, we realize that due to differences in mass, velocity, and acceleration, their time periods and distances from the Sun vary. This explains why each planet follows a unique path and does not collide with others — as they are bound by their individual gravitational interactions. The simulation also helped us understand how changes in mass impact planetary motion.

Although this model is simple, it still provides a clear understanding of how planetary revolution works and how physical and astronomical principles can be visualized through simulation. It gives us insight into how physics explains real-world behavior of celestial bodies.

Limitations:-

- The Euler method used here is not very accurate for long-term simulations, as it introduces small errors over time. More accurate methods like Runge-Kutta are recommended.
- Only the Sun applies gravitational force in this model. In reality, planets also exert gravitational force on each other.
- The Sun is stationary in our model, but in real life, it too moves slightly due to the gravitational pull of the planets.
- We did not include other planets or natural satellites like moons, which exist in the real solar system.
- No 3D representation — the model is a 2D approximation.
- Assumptions were made for values like mass and velocity, without using actual astronomical units.

Future Scope:-

- Add more planets and simulate the effect of their gravitational pull on each other (e.g., Earth–Jupiter interactions).
- Implement adaptive time-stepping, where the simulation automatically changes the time step based on how fast objects are moving — smaller steps for faster motion, larger steps for slower motion.
- Simulate the entire solar system with all 8 planets and their interactions.

- Include planetary rotation and show additional phenomena like day-night cycles.
- Allow the Sun to move in response to planetary forces for greater realism.

Personal Understanding :-

From this project, I understood how simple models based on constant forces and velocity can help explain complex systems like planetary motion. I now understand why some planets are close to the Sun while others are far, and how revolution time changes with distance and mass.

I learned that planets do not fall into the Sun because they have initial velocity that balances gravitational pull, creating stable orbits. Without that velocity, they would collapse into the Sun. Similarly, without proper force balance, they may escape the solar system.

This simulation also helped me understand how orbits are naturally elliptical or circular based on physical laws, without us needing to manually create those shapes. I understood how mass plays a crucial role — with the Sun carrying 99% of the mass, it remains at the center, and other planets revolve around it.

By experimenting with different initial velocities for Earth, I saw how orbital shapes change. This reinforced the idea that orbital behavior emerges directly from gravitational force and motion, not from any pre-written programming logic.
